

When Do Depth Priors Help Sparse-View 3D Gaussian Splatting?

A Controlled Study of Appearance, Geometry, and Prior Corruption

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Abstract

Monocular depth can regularize sparse-view 3D Gaussian Splatting (3DGS), but an inaccurate prior may also constrain reconstruction toward the wrong geometry. We study this interaction through 257 completed runs spanning initialization, view count, depth-loss family and weight, controlled corruption, and Depth Anything V2. On NeRF-Synthetic, clean dense-reference initialization substantially improves both held-out PSNR and reference-depth agreement; exact depth rendered from the original Blender scenes confirms the dense reference to within 0.4% on the scenes used for corruption. Corruption can reverse this advantage: in a three-view noise experiment, joint improvement at 3.9% prior AbsRel becomes joint degradation at 7.9%. This is a transition between sampled settings, not a universal error threshold. Error structure also matters: coherent scale errors produce much greater reference-depth error than larger pixelwise noise. Selected experiments exhibit opposing changes in PSNR and geometry, while many corrupted conditions degrade both. Depth Anything V2 yields useful synthetic results with reference-assisted calibration but poor results at the tested sparse-SfM calibration operating point. On two real DTU scans, its depth initialization reduces structured-light Chamfer error by 45% relative to random initialization at three views. Adding reference-depth supervision further improves DTU geometry, although PSNR, SSIM, and LPIPS do not always agree on appearance. The practical lesson is to assess calibration, intervention path, and geometry alongside multiple image-quality metrics before trusting a depth prior.

1 Introduction

3D Gaussian Splatting (3DGS) represents a scene with explicit, differentiable Gaussian primitives and can synthesize high-quality novel views efficiently [1]. With only a few training views, however, image reconstruction alone leaves substantial geometric ambiguity. A set of primitives may fit the observed images while placing surfaces incorrectly or producing unstable depth in unseen views. Monocular depth offers a dense additional constraint, either by supplying initial 3D points or by supervising rendered depth during optimization.

The usefulness of this constraint depends on what the prior gets wrong. Local noise, a coherent scale error, and image-dependent scale or shift can have different effects even at similar average depth error. Moreover, an image-quality gain does not establish geometric improvement. This distinction matters when a reconstruction will be used for measurement or downstream 3D processing rather than only novel-view rendering.

We investigate three questions. First, how do initialization and the number of views affect the value of clean depth? Second, how do the intervention path, depth loss, and corruption structure change the result? Third, do controlled synthetic observations extend to a real monocular estimator and to independently measured real-scene geometry? We address these questions with a staged experiment over 257 completed runs, using NeRF-Synthetic for controlled reference-depth perturbations and DTU structured-light scans for external geometric evaluation.

Our contribution is a controlled diagnostic study rather than a new rendering method. It separates initialization from continued depth supervision, compares three depth losses, and relates prior error to both appearance and geometry. The results show strong clean-prior benefits, substantial joint failures under corrupted priors, and concrete cases of PSNR–geometry divergence. They also show why a single appearance score or a universal prior-error cutoff is an inadequate basis for choosing depth supervision.

2 Related Work

Gaussian splatting and initialization. Kerbl et al. [1] introduced anisotropic 3D Gaussians, differentiable splatting, and adaptive density control for real-time radiance-field rendering. The initial point distribution is consequential because optimization and densification start from it. Foroutan et al. [2] study alternatives to SfM initialization, including random initialization and structure distilled from NeRF, and find that carefully designed random initialization can approach SfM. This motivates treating initialization as an experimental variable rather than a fixed preprocessing choice. Here we examine it under sparse supervision and explicitly measure depth agreement in addition to image quality.

Depth-regularized sparse-view reconstruction. DNGaussian [3] uses hard and soft depth regularization with global-local normalization. It explicitly discusses conflicts between coarse monocular depth and detailed appearance. Our study complements such methods by holding the reconstruction framework fixed while varying the quality and entry point of depth information. We do not claim to introduce the appearance-geometry conflict; we characterize it with controlled corruption and paired measurements.

Monocular depth and geometric evaluation. Depth Anything V2 [4] provides a strong pretrained relative-depth estimator, but its relative predictions require calibration before use as camera-space depth. We use it as one empirical operating point alongside synthetic perturbations. NeRF-Synthetic [5] supplies calibrated rendered scenes for the controlled study. DTU [6] provides calibrated real images and structured-light reference geometry, enabling a complementary test beyond agreement with a learned dense reconstruction.

3 Method

Let $\{I_i, K_i, T_i\}_{i=1}^V$ be the training RGB images, camera intrinsics, and camera-to-world transforms. Camera poses are held fixed. A depth prior P_i is associated with each training image when depth is used. We distinguish the prior from the reconstruction’s rendered depth $\hat{D}_i$ and from the evaluation reference D_i^{ref} .

Representation and rendering. The scene is a set of Gaussians $\mathcal{G} = \{\mu_j, \Sigma_j, o_j, c_j\}_{j=1}^N$, with center $\mu_j \in \mathbb{R}^3$, positive-definite covariance Σ_j , opacity o_j , and color coefficients c_j . Covariance is parameterized by rotation and positive scales. Projection produces a 2D Gaussian footprint whose opacity contribution at pixel u is $\alpha_j(u)$. For primitives sorted front to back, define

$$T_j(u) = \prod_{k < j} (1 - \alpha_k(u)), \quad w_j(u) = T_j(u) \alpha_j(u). \quad (1)$$

The rendered RGB and expected depth are

$$\hat{I}(u) = \sum_j w_j(u) c_j + \left(1 - \sum_j w_j(u)\right) c_{\text{bg}}, \quad \hat{D}(u) = \frac{\sum_j w_j(u) z_j}{\sum_j w_j(u) + \epsilon}, \quad (2)$$

where z_j is camera-space depth and ϵ stabilizes division. Expected depth is an opacity-weighted rendering statistic; it is not itself a mesh or a guarantee of surface accuracy. We optimize Gaussian parameters using differentiable rasterization and adaptive densification and pruning.

3.1 Depth-prior intervention paths

Initialization. Random initialization samples points within the configured scene extent. SfM initialization uses known-pose triangulation: SIFT features are matched with a ratio test, checked against the known epipolar geometry, and triangulated with DLT. This isolates point initialization without conflating it with camera-pose estimation. Depth initialization backprojects valid training pixels. Writing $T_i = (R_i, t_i)$ and $\tilde{u} = (u_x, u_y, 1)^\top$,

$$X_i(u) = R_i(P_i(u) K_i^{-1} \tilde{u}) + t_i. \quad (3)$$

The resulting colored points seed Gaussian centers. Backprojection requires positive depth in the camera coordinate system; affine-ambiguous relative predictions cannot be substituted directly as metric depth.

Calibrating Depth Anything V2. The model predicts relative inverse depth $d_i(u)$. We calibrate it per image with an affine map in inverse-depth space and then invert, $P_i(u) = 1/(s_i d_i(u) + t_i)$. The reference-assisted (oracle) condition fits (s_i, t_i) by least squares to $1/D_i^{\text{ref}}$ over valid pixels. The SfM-assisted condition fits it to $1/z$ at the triangulated points that view i helped triangulate, widened to all triangulated points projecting into the image when fewer than five are available. With three or more points the fit is least squares followed by one outlier-rejection pass (residuals beyond 3×1.4826 median absolute deviations); with two points it is solved exactly; with a single point, or when the fit yields a non-positive scale, only the scale is fitted ($t_i = 0$). The calibrated prior is restricted to the foreground mask where one exists.

We evaluate three intervention paths (Table 1). Initialization only tests whether the starting geometry remains helpful after RGB optimization. Loss only asks whether depth can guide a reconstruction initialized without dense depth. Initialization plus loss tests the combined effect. A corrupted prior can therefore affect the reconstruction even at $\lambda = 0$ if it supplied the initial point cloud.

Table 1: Depth intervention paths. The loss-only path uses random or SfM initialization; each comparison specifies its baseline.

Path	Initial Gaussian centers	Depth loss during training
Initialization only	Backprojected prior	None ($\lambda = 0$)
Loss only	Random or SfM points	$\lambda > 0$
Initialization + loss	Backprojected prior	$\lambda > 0$

3.2 Training objective and depth losses

The training objective averages image and depth losses over sampled training views:

$$\mathcal{L} = \mathcal{L}_{\text{RGB}} + \lambda \mathcal{L}_{\text{depth}}, \quad \mathcal{L}_{\text{RGB}} = 0.8 \|\hat{I} - I\|_1 + 0.2(1 - \text{SSIM}(\hat{I}, I)), \quad (4)$$

where the L_1 term is averaged over pixels and color channels. Depth terms are computed over valid prior pixels Ω , with $n = |\Omega|$. We compare the following three losses.

Scale-and-shift alignment loss (SSI). The implementation first fits a least-squares affine transformation of rendered depth to the target, then evaluates a mean absolute residual:

$$(s^*, t^*) = \arg \min_{s, t} \sum_{u \in \Omega} (s \hat{D}(u) + t - P(u))^2, \quad (5)$$

$$\mathcal{L}_{\text{SSI}} = \frac{1}{n} \sum_{u \in \Omega} |s^* \hat{D}(u) + t^* - P(u)|. \quad (6)$$

For nondegenerate depth variation, the least-squares solution is $s^* = \sum_u (\hat{D}(u) - \bar{\hat{D}})(P(u) - \bar{P}) / \sum_u (\hat{D}(u) - \bar{\hat{D}})^2$ and $t^* = \bar{P} - s^* \bar{\hat{D}}$, with numerical stabilization in code. Importantly, fitting uses squared residuals while the final penalty is L_1 .

Pearson loss. With $x_u = \hat{D}(u) - \bar{\hat{D}}$ and $y_u = P(u) - \bar{P}$,

$$\mathcal{L}_{\text{Pearson}} = 1 - \frac{\sum_u x_u y_u}{\sqrt{\sum_u x_u^2} \sqrt{\sum_u y_u^2 + \epsilon}}. \quad (7)$$

This encourages correlated spatial variation without imposing an absolute scale or offset. Positive affine changes of the target leave correlation unchanged in the nondegenerate, unstabilized case.

Absolute depth loss. For a prior calibrated into the reconstruction’s depth coordinates,

$$\mathcal{L}_{\text{abs}} = \frac{1}{n} \sum_{u \in \Omega} |\hat{D}(u) - P(u)|. \quad (8)$$

Unlike the aligned losses, this directly penalizes scale and offset disagreement. It can anchor correct depth, but it can also impose an incorrect metric target.

The losses differ in units and normalization, so equal numerical λ values do not imply equal regularization strength. In particular, the implemented SSI loss removes an affine mismatch through fitting, but is not invariant in magnitude to scaling its target. Ideally, for $a > 0$ and unchanged valid pixels,

$$\mathcal{L}_{\text{SSI}}(\widehat{D}, aP + b) = a \mathcal{L}_{\text{SSI}}(\widehat{D}, P). \quad (9)$$

Thus target scaling can change the effective loss weight even when the affine mismatch is fitted away.

3.3 Controlled prior corruption

We perturb the dense reconstruction reference while keeping the RGB images and camera poses fixed. The families are

$$\text{zero-mean noise: } P_i(u) = D_i^{\text{ref}}(u) + \eta_i(u), \quad \mathbb{E}[\eta_i(u)] = 0, \quad (10)$$

$$\text{global affine: } P_i(u) = a D_i^{\text{ref}}(u) + b, \quad (11)$$

$$\text{per-image affine: } P_i(u) = a_i D_i^{\text{ref}}(u) + b_i. \quad (12)$$

The noise control varies perturbation intensity; global affine corruption shares parameters across images, whereas per-image affine corruption allows inconsistent calibration across views. The highlighted global conditions are pure scale changes, $b = 0$. Noise intensity and affine jitter are nominal experiment controls; we compare their realized severity through measured prior AbsRel rather than treating those controls as interchangeable percentages. Corruptions are applied to prior depth, not to the evaluation reference.

The staged grid avoids redundant configurations. At $\lambda = 0$, changing the unused loss family cannot create a new intervention. Positive global affine changes are invisible to ideal Pearson correlation in a loss-only path. For SSI, equivalence additionally requires accounting for the effective weight in Eq. (9). Neither argument removes affine-corrupted depth initialization, which changes the actual backprojected points. Omitted combinations are therefore design exclusions, not evidence of successful robustness.

3.4 Evaluation metrics and matched comparisons

Appearance. We evaluate held-out RGB images with PSNR, SSIM, and LPIPS. PSNR measures pixel error, $\text{PSNR} = -10 \log_{10}(\text{MSE})$ for images in $[0, 1]$; higher is better. SSIM measures local structural similarity and is also better when higher. LPIPS measures learned perceptual distance and is better when lower. We report all three where numerical results are available and describe PSNR-specific findings as such.

Reference-depth agreement. The principal synthetic reconstruction metric is unaligned mean absolute depth error. Prior quality is measured separately:

$$\text{MAE}(\widehat{D}, D^{\text{ref}}) = \frac{1}{n} \sum_{u \in \Omega} |\widehat{D}(u) - D^{\text{ref}}(u)|, \quad \text{AbsRel}(P, D^{\text{ref}}) = \frac{1}{n} \sum_{u \in \Omega} \frac{|P(u) - D^{\text{ref}}(u)|}{D^{\text{ref}}(u) + \epsilon}. \quad (13)$$

MAE is in scene depth units; AbsRel is dimensionless. Prior AbsRel is an input diagnostic, not the output reconstruction error. The evaluation pipeline also computes RMSE and thresholded depth ratios. We emphasize unaligned MAE because an affine fit before evaluation can remove the very error being studied: a depth map with the correct shape but the wrong global scale could score almost perfectly after alignment.

Independent DTU geometry. For reconstructed points Q and structured-light points G , accuracy averages nearest-neighbor distances from Q to G , and completeness averages distances in the reverse direction. The reported DTU score is their mean,

$$C_{\text{DTU}} = \frac{1}{2} \left[\frac{1}{|Q|} \sum_{q \in Q} \min_{g \in G} \|q - g\|_2 + \frac{1}{|G|} \sum_{g \in G} \min_{q \in Q} \|g - q\|_2 \right], \quad (14)$$

after the evaluation protocol’s spatial filtering. Distances are in millimeters, without a post-hoc scale-and-shift fit. This is a different measurement from synthetic reference-depth MAE.

Comparisons. For a geometry error E , we use $\Delta E = 100(E/E_0 - 1)$ and $\Delta \text{PSNR} = \text{PSNR} - \text{PSNR}_0$. Each baseline matches the dataset, scene, view count, split, and training setup. Intervention comparisons use random initialization without depth unless stated otherwise; loss ablations hold initialization fixed and compare $\lambda > 0$ against $\lambda = 0$. Scene averages use the same scenes in both arms. Opposing metric directions establish a particular divergence, not its frequency across an unbalanced experimental grid.

4 Experimental Setup

4.1 NeRF-Synthetic and the dense reconstruction reference

The main controlled study uses *Lego*, *Chair*, and *Ship*, with 3, 5, or 9 training views. We use a fixed farthest-view selection for these comparisons, render at 400×400 with a white background, train for 7,000 iterations, and evaluate 25 held-out views. The main study uses spherical-harmonic degree zero, restricting view-dependent color as an alternative way to explain sparse observations. The reproduction experiment instead uses degree three and the published eight-view DietNeRF split.

For each controlled scene, we fit a separate dense 3DGS reconstruction using all 400 images at 800×800 , with alpha supervision, and render its expected depth as D^{ref} . Clean-reference interventions are therefore *oracle conditions*: their depth is derived from substantially more information than the sparse training views, including views reserved for sparse-model evaluation. They measure the potential value of strong geometric information, not the performance of a deployable sparse-view estimator.

Table 2 assesses this reference in two ways. Multi-view consistency and depth-rank agreement with the released depth images are internal checks: a reconstruction can be internally consistent and still systematically inaccurate. We therefore also rendered exact depth from the modified Blender scenes the dataset was created from, which are not part of the release but are available through the NeRF issue tracker,¹ at the 25 evaluation views and at the 16–17 views any split uses as input. These renders reproduce the released geometry: their silhouettes match the released alpha masks at $\text{IoU} \geq 0.997$, and where exact and reference depth disagree by more than 3%, the released 8-bit depth images agree with the exact depth on 95–99.9% of those pixels.

Against exact depth, the reference has 0.22% AbsRel on *Lego* and 0.36–0.38% on *Chair*, but 2.3–2.4% on *Ship*, where on about a quarter of the pixels it lies a median 5.7% in front of the water surface (Figure 1); multi-view consistency had put *Ship* at 0.17%, because a surface displaced consistently in every view is self-consistent. The corruption and Depth Anything V2 experiments use only *Lego* and *Chair*. Recomputing every corrupted prior against exact depth instead of the reference changes no prior AbsRel by more than 0.25 percentage points; the smallest noise level moves from 3.94% to 3.97%. Section 5.8 examines which geometry comparisons the reference error could affect.

Table 2: Dense-reference quality. Left: internal consistency checks, the median relative reprojection disagreement between neighboring views and rank correlation with the released depth PNGs. Right: agreement with exact depth rendered from the Blender scenes; AbsRel is computed as prior error is $(400 \times 400, \text{unaligned})$, on the evaluation views and on the input views; the within-1% fraction is at full resolution; silhouette IoU compares Blender coverage with the released alpha masks.

Scene	Internal consistency		Against exact Blender depth		
	Reprojection	Rank corr.	AbsRel (eval / input)	Within 1%	Silhouette IoU
Lego	0.043%	0.987	0.22% / 0.22%	94.8%	0.9996
Chair	0.049%	0.990	0.36% / 0.38%	90.8%	0.9997
Ship	0.171%	0.961	2.33% / 2.39%	48.3%	0.9974

4.2 DTU

We evaluate scans 1 and 6 from the DTU SampleSet with 3 or 9 training views at 400×300 . Cameras use the provided calibration. Structured-light geometry supplies the independent evaluation target and, in oracle conditions, depth projected into the training cameras. Evaluation uses 0.2 mm point downsampling, the observability mask and ground-plane filtering, and a 20 mm distance cutoff. We report the mean of accuracy and completeness. Because the appearance evaluation uses full images without object masks, these RGB scores should not be compared directly with masked DTU numbers from other papers.

¹<https://github.com/bmild/nerf>, issues 59 and 198; we used the community copy `blend_files_fixed.zip`.

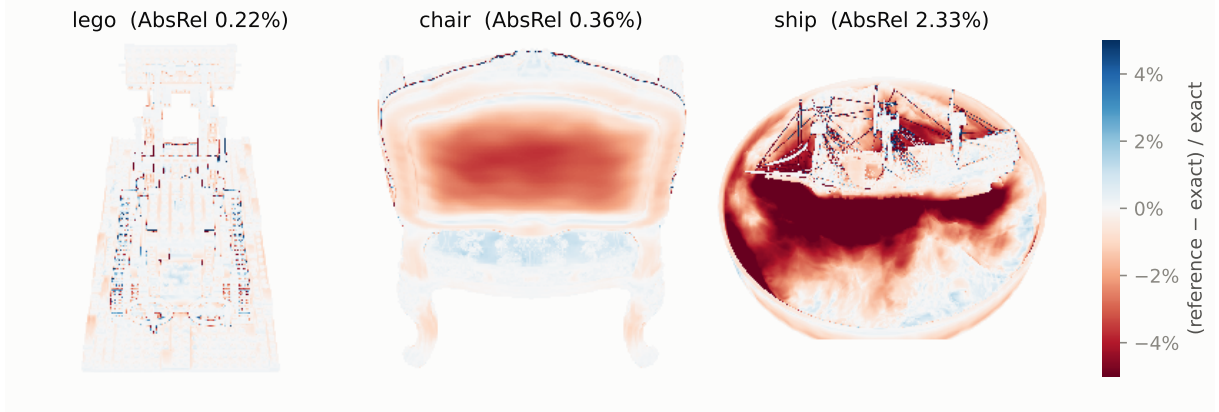


Figure 1: Signed relative error of the dense reference against exact Blender depth on the first evaluation view (red: reference in front of the true surface; blue: behind). Lego is exact up to edges; Chair’s reference lies 2–4% in front of the upholstered back; Ship’s lies 5% and more in front of the water.

4.3 Implementation and reproducibility

Our implementation uses PyTorch and the installed third-party gsplat 1.5.3 rasterizer [7]. The authors’ code is under `src/`, `scripts/`, and `tests/`; gsplat is a dependency rather than an edited copy of the rasterizer. Dependencies are specified in `requirements.txt` and `requirements-gpu.txt`. Training uses seeded, automated experiment stages and supports resuming sweeps. The project repository and executed Colab notebook provide the setup, training, evaluation, and analysis workflow, including the exact-depth validation and the scripts that draw the quantitative figures.²

We corrected two integration issues in our code while retaining the rasterizer as an unmodified installed dependency. We made the packed/unpacked settings of `rasterization()` and `DefaultStrategy` consistent, correcting the attribution of densification gradients to Gaussians. We also explicitly perform the vanilla 3DGS opacity reset, correcting an operator-precedence error that prevented the intended reset condition from firing. These corrections materially improved reconstruction quality: on Ship, PSNR increased by approximately 1.7 dB and the floater fraction decreased from about 45% to 16%.

The study contains 257 completed runs across the stages in Table 3. These runs span different experimental settings; they are not 257 independent repeated-seed trials. The grid is staged rather than fully Cartesian: broad initialization and view-count comparisons are followed by loss and corruption studies and real-scene validation. This design allocates computation to the study’s central questions while retaining matched comparisons within each stage.

Table 3: Experimental breadth. Counts describe completed runs in the staged study; they are not sample sizes for repeated-seed significance tests.

Stage	Runs	Stage	Runs
Eight-scene reproduction	8	Global affine corruption	32
Initialization $\times$ views	27	Per-image affine corruption	32
Initialization/loss combinations	18	Synthetic real-model experiment	20
Loss-family/ λ sweep	48	DTU baselines	8
Noise corruption	40	DTU reference-depth ablation	8
		DTU real-model runs	16
		Total	257

5 Results

5.1 Reproduction sanity check

Before interpreting depth interventions, we test the RGB-only implementation against the 3DGS baseline reported by DNGaussian [3]. Under the eight-view NeRF-Synthetic protocol, our mean scores are 21.27 dB

²<https://github.com/AMEER7525/depth-prior-tax>

PSNR, 0.853 SSIM, and 0.149 LPIPS (Table 4). The corresponding published scores are 22.23, 0.858, and 0.114. The 0.96 dB PSNR gap meets our approximate one-decibel sanity criterion, while the SSIM and LPIPS gaps show that this is not an exact reproduction. Accordingly, subsequent causal comparisons use our own matched baselines rather than attributing differences from published systems to depth alone.

Table 4: RGB-only reproduction: all eight NeRF-Synthetic scenes, eight-view DietNeRF split, 400×400 , 6,000 training iterations, random initialization, SH degree three. The published comparison is DNGaussian’s *3DGS baseline row*, not DNGaussian itself.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Our 3DGS implementation	21.27	0.853	0.149
Published 3DGS baseline [3]	22.23	0.858	0.114

5.2 Initialization versus number of views

To isolate initialization, we disable the depth loss and compare reference-depth, SfM, and random initialization at each view count. Table 5 and Figure 2 show that clean reference depth improves both PSNR and reference-depth MAE at all three counts. At three views it raises PSNR from 15.00 to 19.43 dB and reduces MAE from 0.227 to 0.068, a 69.8% reduction relative to random initialization. At nine views the corresponding values are 23.65 to 27.14 dB and 0.084 to 0.026, a 69.5% reduction.

Table 5: Initialization only ($\lambda = 0$), averaged over Lego, Chair, and Ship. Entries are PSNR in dB / unaligned reference-depth MAE in scene units.

Initialization	3 views	5 views	9 views
Clean reference depth	19.43 / 0.068	23.14 / 0.040	27.14 / 0.026
SfM	15.51 / 0.254	18.97 / 0.129	24.99 / 0.058
Random	15.00 / 0.227	18.48 / 0.145	23.65 / 0.084

Additional views improve all three initializations. SfM achieves higher PSNR than random initialization at 3, 5, and 9 views, but has lower MAE only at 5 and 9 views. Thus even without a monocular prior, improved image fitting and improved reference-depth agreement need not coincide. These SfM-versus-random orderings depend on the scene mix: averaged over Lego and Chair alone, SfM has lower PSNR than random initialization at 3 and 5 views and lower MAE only at 9 views. The advantage of clean reference depth, by contrast, holds on both measurements at every view count without Ship, whose reference is the least accurate (three views: 20.81 dB / 0.050 against 16.52 dB / 0.198 for random initialization). The clean-depth results establish an oracle benefit of informative initial geometry; they do not imply that a real monocular model supplies equally accurate points.

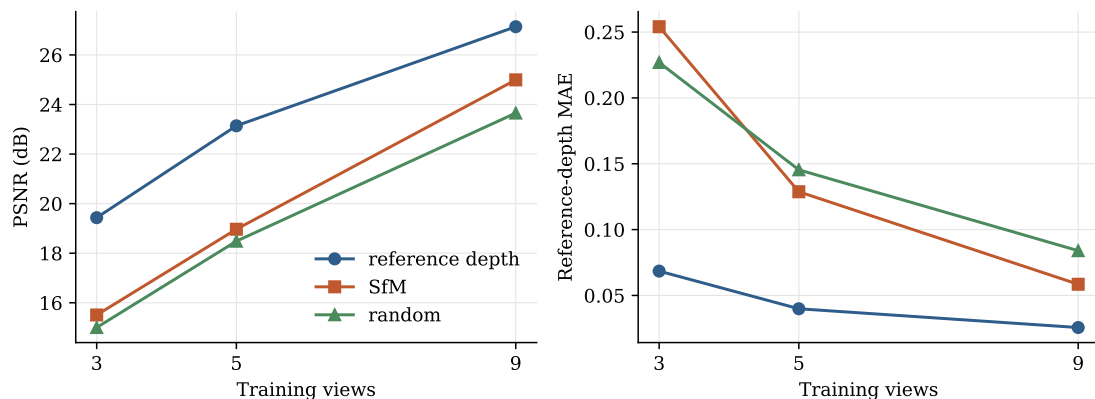


Figure 2: Initialization and view count. Curves connect the three-scene means in Table 5; they do not depict confidence intervals. Clean-reference initialization remains beneficial as view coverage increases.

5.3 Depth-loss weight and loss family

With clean-depth initialization fixed, we sweep SSI, Pearson, and absolute losses over $\lambda \in \{0.03, 0.1, 0.3, 1\}$ at three and nine views on Lego and Chair. Figure 3 shows that the effect depends on both loss family and view count. Pearson deteriorates sharply at the largest weight: the three-view curve ends near 16.7 dB PSNR and 0.27 reference-depth MAE. SSI and absolute loss are less severely affected in this range. At nine views, the SSI curve has lower MAE at the larger tested weights than at 0.1, whereas the Pearson curve deteriorates strongly at 1. These patterns support tuning the loss within a specified view regime rather than transferring one weight universally.

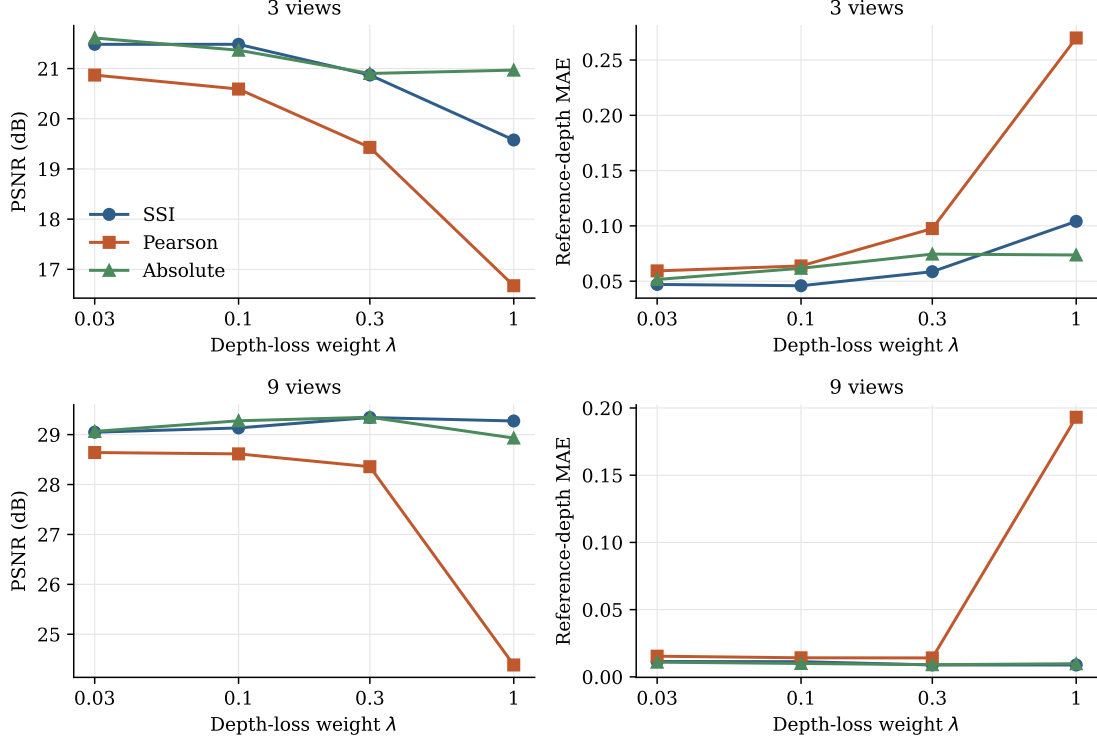


Figure 3: Clean-prior loss sweep with depth initialization at three and nine views. Lines summarize the loss-sweep configurations at each tested weight; no repeated-seed uncertainty is implied. No-loss effects are assessed through the matched scene-level pairs in Table 6.

Matched single-scene comparisons further show why selecting λ by PSNR alone can be misleading (Table 6). On Chair at three views, adding an absolute loss at $\lambda = 0.03$ improves PSNR by 0.95 dB but raises MAE by 26.0%. SSI at the same weight raises PSNR by 0.54 dB and MAE by 12.1%. Conversely, on Lego at nine views, increasing Pearson’s weight from 0.1 to 0.3 reduces MAE by 3.6% while lowering PSNR by 0.31 dB. These opposing changes are small in absolute terms, at most 0.012 in MAE, and lie within the largest shift the reference’s own error could cause (Section 5.8). They show that the two measurements can move apart under a clean target, but not reliably by how much.

Table 6: Selected matched clean-prior loss ablations with depth initialization. Positive Δ MAE means worse reference-depth agreement. These comparisons are individual configurations, not significance estimates.

Scene / views	Loss	λ change	Δ PSNR (dB)	Δ MAE
Chair / 3	Absolute	0 $\rightarrow$ 0.03	+0.95	+26.0%
Chair / 3	SSI	0 $\rightarrow$ 0.03	+0.54	+12.1%
Lego / 9	Pearson	0.1 $\rightarrow$ 0.3	−0.31	−3.6%
Lego / 9	Pearson	0.3 $\rightarrow$ 1	−0.21	−2.3%

We use SSI at $\lambda = 0.1$ for the highlighted corruption comparison as a moderate tested setting. This is an experimental operating choice, not a demonstrated universal optimum. Since the losses have different scales (Section 3.2), their weight curves compare implemented objectives, not equally strong constraints.

5.4 Controlled prior degradation

To test when the oracle advantage survives imperfect depth, we vary corruption structure and severity while holding the reconstruction setting fixed. Table 7 reports the three-view, depth-initialization-plus-SSI comparison on Lego and Chair. Its matched random-initialization baseline is 16.52 dB PSNR and 0.198 MAE; the three-scene mean from Table 5 is not used for this comparison.

Noise: a transition between sampled operating points. At 3.9% prior AbsRel, PSNR is 18.32 dB and MAE is 0.122: a 1.80 dB gain and a 38.3% error reduction. At 7.9%, PSNR falls to 16.28 dB and MAE rises to 0.258, reversing both signs relative to the matched baseline. The 15.7% condition is worse still, at 14.42 dB and 0.365 MAE: PSNR decreases by 2.11 dB and depth error increases by 84.5%. This locates an empirical transition between the sampled 3.9% and 7.9% conditions. Its location is specific to the scenes, intervention path, loss, and view setting tested here.

Coherent and image-dependent affine error. A global scale factor of 0.8 produces 20% prior AbsRel and yields 11.97 dB PSNR with 0.859 MAE. Its depth error is more than twice that of the 30.9% noise condition (0.391), despite the smaller prior AbsRel. The factor-1.5 condition is worse, reaching 1.705 MAE. Across the tested global-affine family, all 32 runs spanning scales and intervention paths showed joint degradation relative to their baselines. This family-level result, together with the matched examples, suggests that coherent calibration errors can be particularly damaging.

Per-image affine corruption also changes character with severity. The displayed 6.2% condition improves both measurements, whereas the 25.6% condition reaches 13.57 dB and 0.596 MAE. These comparisons indicate that error structure matters in addition to its size. They are not a fully error-matched factorial test: the corruption families were sampled at different realized AbsRel values, so the results do not isolate a universal causal coefficient for structure.

Table 7: Controlled corruption on Lego and Chair, three views, depth initialization plus SSI loss at $\lambda = 0.1$. The baseline uses the same two scenes and view setup, with random initialization and no depth loss. MAE measures rendered depth against the dense reference; prior AbsRel measures the corrupted input.

Corruption	Control	Prior AbsRel	PSNR (dB)	Reference MAE
Matched baseline	No prior	—	16.52	0.198
Noise	$\sigma = 0.05$	3.9%	18.32	0.122
	$\sigma = 0.10$	7.9%	16.28	0.258
	$\sigma = 0.20$	15.7%	14.42	0.365
	$\sigma = 0.40$	30.9%	12.01	0.391
Global affine	$a = 0.8$	20.0%	11.97	0.859
	$a = 1.5$	50.0%	10.33	1.705
Per-image affine	Jitter 0.05	6.2%	18.12	0.125
	Jitter 0.20	25.6%	13.57	0.596

How the prior enters matters. Figure 5 retains the broader comparison of initialization only, loss only, and initialization plus loss for SSI and absolute supervision. Errors introduced by backprojection remain consequential even when the loss fits away affine disagreement. Loss-only SSI has a different sensitivity, while absolute supervision directly exposes a miscalibrated target to optimization. The distinction is visible in individual outcomes: a Lego initialization-only per-image-affine prior at 6.2% AbsRel raises PSNR by 0.72 dB but raises MAE by 51.8%; a random-initialized SSI loss-only noise prior at 7.8% AbsRel lowers PSNR by 0.42 dB while reducing MAE by 10.7%. Thus similar prior-error magnitudes do not determine either the direction of change or the mechanism of failure.

5.5 Depth Anything V2

The controlled perturbations let us ask whether a real predictor operates in a useful regime. We evaluate the large Depth Anything V2 model under reference-assisted oracle calibration and sparse-SfM-assisted calibration (Section 3.1). Table 8 summarizes the synthetic three-view results. With reference-assisted calibration, prior AbsRel is 1.8%, and the depth-initialized configurations reach 18.2–18.8 dB PSNR with MAE 0.085–0.115. Under the tested SfM-assisted pipeline, prior AbsRel is 24.7%, with 13.1–13.7 dB

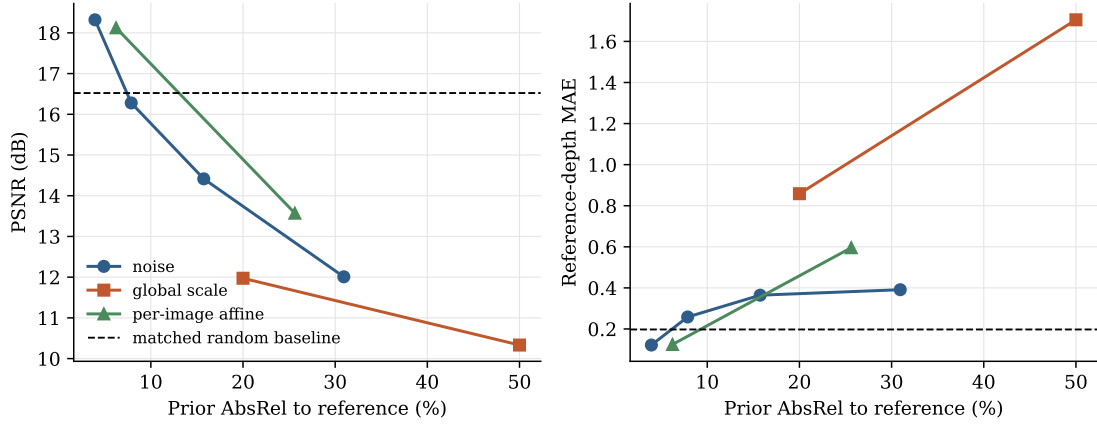


Figure 4: Matched corruption comparison from Table 7. Dashed lines show the two-scene no-prior baseline. Curves connect sampled conditions and do not estimate a continuous threshold. Equal horizontal positions would mean equal prior AbsRel, not equivalent corruption structure.

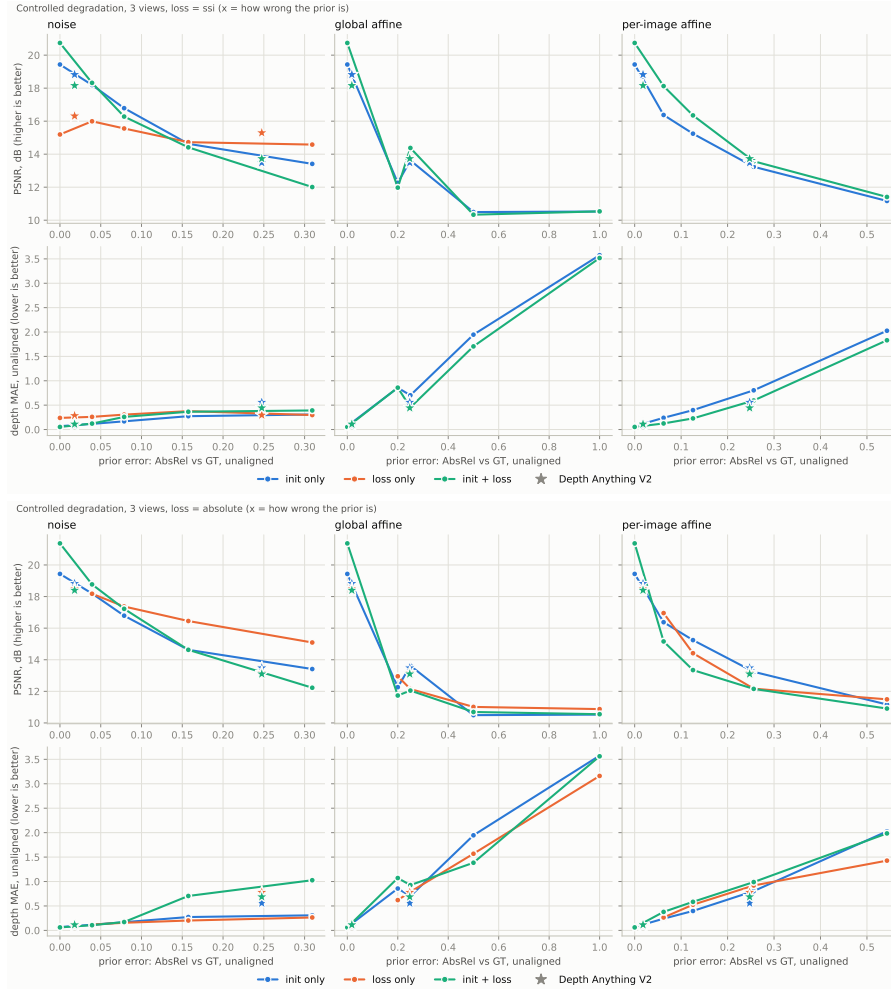


Figure 5: Three-view corruption curves across intervention paths: SSI supervision (top) and absolute supervision (bottom). Each block plots PSNR above unaligned reference-depth MAE, against prior AbsRel. Colors distinguish initialization only, loss only, and initialization plus loss; stars show the real-model operating points. Missing path/family combinations reflect the staged design and pruning, not measured zero effect. Curves interpolate displayed configurations, without uncertainty bands.

PSNR and MAE 0.44–0.69. The former is useful relative to the two-scene no-prior baseline; the latter degrades both measurements.

An additional per-image affine diagnostic reduces the SfM-calibrated prior’s residual AbsRel to 3.5%. This indicates a substantial affine component in the depth disagreement, but does not establish which part of the calibration caused it. The calibration is performed in inverse-depth space before inversion, and the SfM fit uses 6–7 points per view on average. The small number of synthetic SfM correspondences is a plausible contributor, but was not varied independently and cannot by itself explain the entire difference.

Table 8: Depth Anything V2 Large on the synthetic three-view setting. PSNR and MAE intervals span the three depth-initialized configurations under each calibration condition (initialization only, and with SSI or absolute loss at $\lambda = 0.1$); they are not confidence intervals. The loss-only configuration with random initialization falls outside these intervals (16.3 dB / 0.284 oracle; 15.3 dB / 0.296 SfM-assisted). Oracle calibration uses reference information unavailable in a normal sparse-view reconstruction.

Calibration condition	Prior AbsRel	PSNR (dB)	Reference MAE
Reference-assisted (oracle)	1.8%	18.2–18.8	0.085–0.115
Three-view SfM-assisted	24.7%	13.1–13.7	0.44–0.69

Figure 6 illustrates the consequence on a held-out Lego view. Clean-reference initialization gives more coherent RGB and rendered depth than the random baseline. A scale-biased initialization disrupts both, and the tested SfM-calibrated real-model configuration also produces a degraded reconstruction. The depth panels visualize spatial structure; without a shared numerical color scale, color differences should not be read as calibrated depth-error measurements.

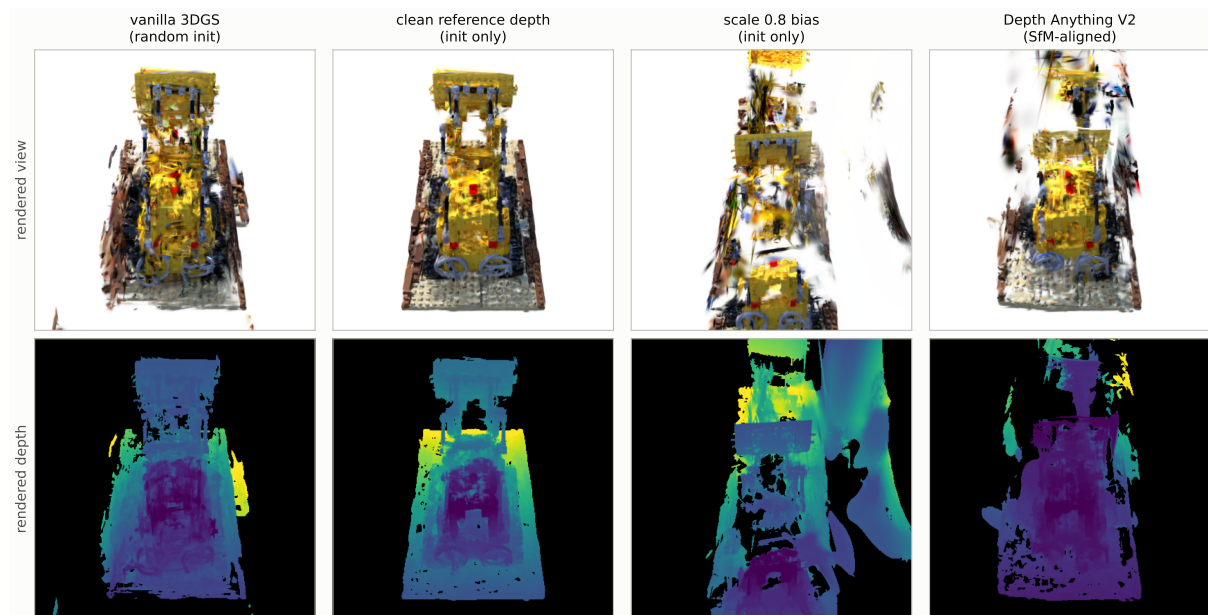


Figure 6: Qualitative three-view Lego comparison at the same held-out camera. These are rendered RGB images and expected-depth maps, not ground-truth surface maps. The comparison illustrates reconstruction behavior under clean, corrupted, and real-model priors; quantitative conclusions use the corresponding evaluation metrics.

5.6 DTU real-scene validation

We next ask whether depth benefits extend to geometry measured independently of a learned reconstruction. Table 9 compares the tested pipelines on DTU scans 1 and 6. At three views, Depth Anything V2 initialization reduces mean Chamfer from the random baseline’s 5.010 mm to 2.750 mm, a 45.1% reduction, while PSNR increases from 8.77 to 12.11 dB. At nine views, the corresponding Chamfer scores are 5.247 and 1.786 mm. The SfM-initialized pipeline with DAv2 depth loss achieves the highest PSNR among the displayed pipelines, reaching 20.43 dB at nine views, while its 1.374 mm Chamfer remains above the clean-reference initialization result of 0.987 mm.

The SfM rows isolate the DAv2 loss for a fixed initialization. At three views, adding it raises PSNR by 2.51 dB, and SSIM and LPIPS also improve (0.428 to 0.513 and 0.509 to 0.444), while Chamfer error grows by 13.3% (3.445 to 3.902 mm). At nine views it improves both PSNR (+1.57 dB) and Chamfer (−16.1%). The remaining rows compare complete pipelines; they do not isolate initialization from regularization or prove that correspondence count alone explains the contrast with synthetic scenes.

Table 9: DTU pipeline comparison, means over scans 1 and 6. Entries are full-image PSNR in dB / structured-light Chamfer in mm. The structured-light reference and DAv2 depth-initialization rows use initialization only ($\lambda = 0$); the SfM + DAv2 row adds the SSI depth loss at $\lambda = 0.1$ to the SfM initialization of the row below it.

Pipeline	3 views	9 views
Structured-light reference initialization	11.73 / 1.218	17.33 / 0.987
DAv2 depth initialization	12.11 / 2.750	15.99 / 1.786
SfM initialization + DAv2 depth loss	13.47 / 3.901	20.43 / 1.374
SfM initialization, no depth	10.96 / 3.445	18.86 / 1.639
Random initialization, no depth	8.77 / 5.010	9.70 / 5.247

A matched reference-depth loss ablation. To isolate continued depth supervision, we keep structured-light reference initialization fixed and compare $\lambda = 0$ with SSI at $\lambda = 0.1$. Table 10 reports all three appearance metrics alongside Chamfer. At three views, mean Chamfer decreases from 1.2184 to 0.8042 mm, a 34.0% reduction, while PSNR decreases from 11.73 to 11.37 dB. However, SSIM increases from 0.5999 to 0.6147 and LPIPS decreases from 0.3538 to 0.3411. Thus geometry improves while PSNR worsens, but both other appearance metrics improve. Calling this simply a loss of appearance quality would obscure the actual result.

At nine views, mean Chamfer decreases from 0.9867 to 0.6587 mm, a 33.2% reduction. Mean PSNR and SSIM decrease, and LPIPS increases slightly. Both scans improve geometrically at both view counts (Figure 7), but nine-view PSNR is heterogeneous: it increases by 0.30 dB on scan 1 and decreases by 1.19 dB on scan 6. These paired observations support a real geometry benefit while showing why two-scan averages should not be treated as universal behavior.

Table 10: Matched DTU reference-depth loss ablation, means over scans 1 and 6, with identical reference initialization. The three-view loss improves Chamfer, SSIM, and LPIPS despite reducing PSNR. Chamfer means are computed from the per-run accuracy/completeness evaluation.

Views	SSI λ	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Chamfer (mm) $\downarrow$
3	0	11.73	0.5999	0.3538	1.2184
3	0.1	11.37	0.6147	0.3411	0.8042
9	0	17.33	0.8127	0.2093	0.9867
9	0.1	16.88	0.8109	0.2128	0.6587

5.7 Appearance–geometry divergence

The preceding experiments distinguish two outcomes: joint degradation under a bad prior, and opposing changes in image and geometric measurements. Table 11 gives five explicit depth-prior cases of the latter relative to matched random baselines. Three have higher PSNR but worse reference-depth MAE; two have lower PSNR but better MAE. The per-image-affine initialization-only Lego case is particularly clear: PSNR increases by 0.72 dB while MAE increases by 51.8%. The three per-image-affine cases change MAE by 0.048–0.087 in absolute terms, beyond the largest shift the reference’s own error could cause; the two noise cases (−0.015 and −0.023) are not (Section 5.8). The DTU result above is unaffected by the synthetic reference and gives a real-scene case: adding the DAv2 loss to SfM initialization at three views improves all three appearance metrics while Chamfer error grows by 13.3%. The selected loss-weight comparisons and DTU ablation provide additional examples under a different baseline definition.

Figure 8 summarizes the broader configuration outcomes. Large joint degradations occupy the upper-left region, while the upper-right region shows higher PSNR with greater geometric error. The plot illustrates the diversity and scale of responses across the study. Neither these selected examples nor the staged grid provides an unbiased estimate of how frequently divergence occurs. In addition, PSNR

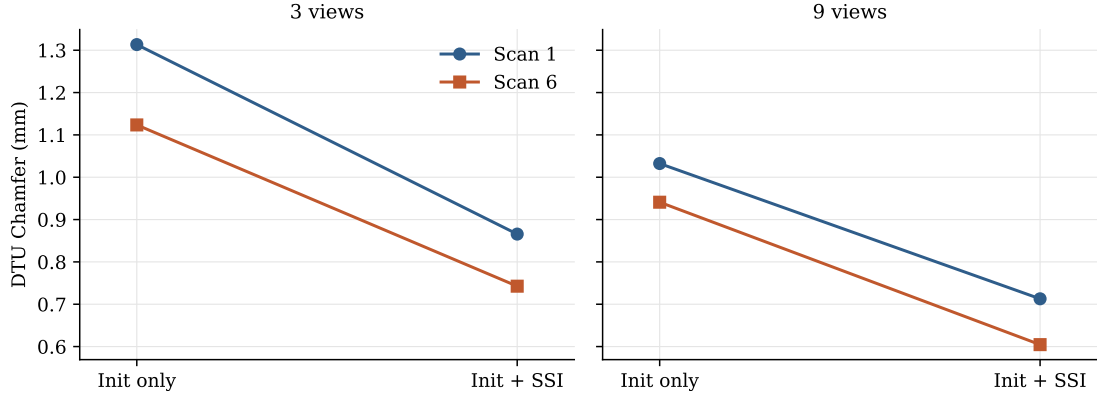


Figure 7: DTU geometry improves on both scans when SSI supervision is added to reference-depth initialization. Lines connect matched configurations for the same scan; they are not error bars or repeated-seed confidence intervals.

Table 11: Selected depth-prior divergences at three views. Each change is relative to that scene’s random-initialization, no-depth baseline. All rows use a depth prior through initialization, a loss, or both. A positive MAE change means worse reference-depth agreement.

Scene	Prior	Path	Loss / λ	Δ PSNR	Δ MAE
Lego	Per-image affine	Init only	— / 0	+0.72	+51.8%
Lego	Per-image affine	Loss only	Absolute / 0.1	+0.69	+34.1%
Lego	Noise	Loss only	SSI / 0.1	−0.42	−10.7%
Chair	Noise	Init only	— / 0	−0.41	−8.9%
Chair	Per-image affine	Loss only	Absolute / 0.1	+0.17	+33.9%

divergence should not be interpreted as disagreement with all appearance metrics, as the three-view DTU results demonstrate.

5.8 What the reference error can change

Appearance metrics and DTU Chamfer do not use the dense reference. For synthetic geometry, a run’s MAE against the reference differs from its MAE against exact depth by at most the reference’s own MAE at the evaluation views, which is 0.008 on Lego, 0.013 on Chair, and 0.093 on Ship; a difference between two runs can therefore shift by at most twice that. The corruption thresholds, the loss-family comparisons, and the Depth Anything V2 operating points rest on Lego and Chair, where the smallest corruption swept (3.9%) is 10–18 times the reference’s AbsRel and the oracle-calibrated Depth Anything V2 prior (1.8%) 5–8 times. Ship enters only the initialization and view-count comparisons: its clean-reference prior is effectively a 2.4%-error prior on the input views, and excluding it leaves the advantage of clean-reference initialization intact (Section 5.2). Under the worst-case bound, the per-image-affine divergences of Table 11 stand, whereas the four clean-prior steps of Table 6 and the two noise divergences of Table 11 are too small to rule out reference error.

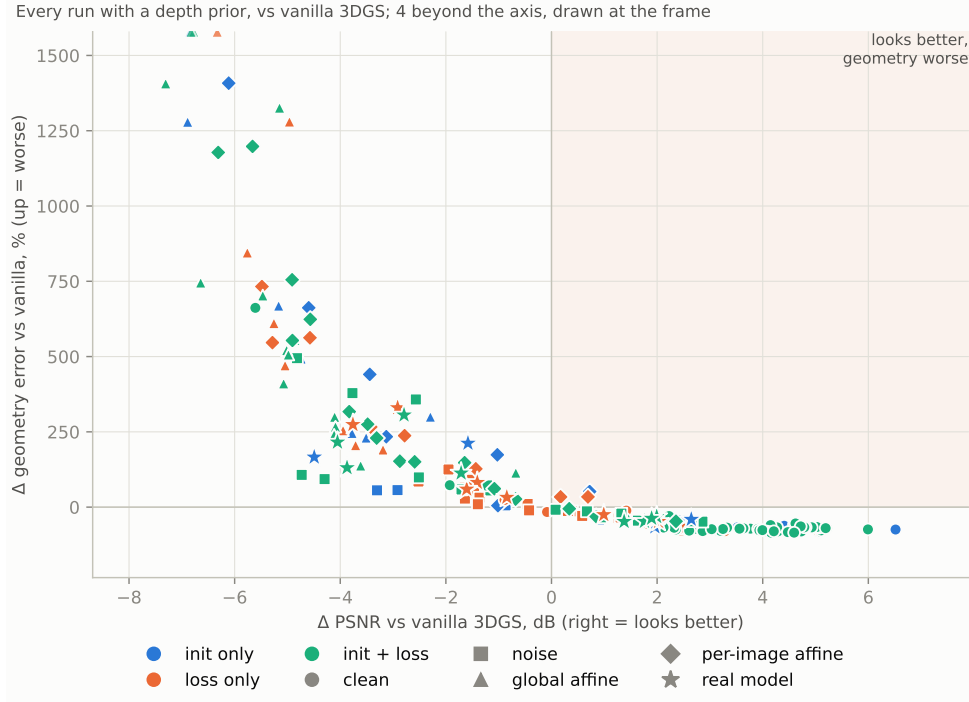


Figure 8: Broad appearance–geometry summary for depth-prior configurations. Horizontal position is PSNR change from the matched no-prior baseline; vertical position is relative geometry-error change (reference-depth MAE on synthetic scenes and Chamfer on DTU). Colors indicate intervention paths and shapes prior families. Four out-of-range points are drawn at the frame, so their boundary positions are clipped. The shaded upper-right region represents higher PSNR with worse geometry. The plot supports qualitative comparison of outcomes, not a divergence-frequency estimate or a common absolute geometry scale.

6 Discussion

The study identifies three distinct decisions behind a successful depth intervention. The first is whether the prior supplies useful initial geometry. Clean-reference initialization is consistently strong in the controlled synthetic experiment, but a coherent scale error can place the starting points so poorly that RGB optimization does not repair them. An invariant loss cannot undo this initialization error merely by ignoring affine disagreement.

The second decision is how strongly to constrain the subsequent optimization. The loss sweep and matched ablations show that more depth supervision is not uniformly better, even with clean depth. SSI, Pearson, and absolute losses impose different constraints, and their numerical weights are not interchangeable. Loss selection should therefore account for calibration quality, view coverage, and the metric relevant to the application.

The third decision concerns what counts as improvement. Synthetic MAE measures agreement with a dense learned reference, which exact Blender depth shows to be accurate on Lego and Chair but not on Ship’s water; DTU Chamfer measures agreement with independently acquired geometry. PSNR, SSIM, and LPIPS also capture different aspects of rendering quality. On DTU at three views, reference-depth supervision improves geometry, structure, and perceptual similarity together while PSNR declines, and a DAv2 loss improves all three appearance metrics while geometry worsens. A single scalar conceals both distinctions.

Practically, we recommend validating prior calibration and cross-view consistency before backprojection, comparing initialization-only and loss-based interventions against matched baselines, and inspecting geometry together with several appearance metrics. The tested noise transition and moderate SSI weight are useful operating observations, not deployment guarantees. The real-model results reinforce this point: the same predictor can be useful or harmful under different calibration and scene conditions.

7 Limitations

The main controlled study covers only three NeRF-Synthetic scenes, with the corruption experiments concentrated on Lego and Chair, and real-scene validation covers only two DTU scans. The synthetic reference is a dense learned reconstruction rather than exact depth; measured against exact Blender depth, it is accurate to 0.4% on Lego and Chair but 2.3% off on Ship, which limits how finely Ship’s geometry numbers can be read. Clean-reference experiments deliberately use oracle information, including dense-view information beyond the sparse training set. Every configuration was trained once, with a single seed, and the staged sweep does not sample all configurations uniformly, so small differences and event frequencies should not be generalized. The real-model comparison does not independently isolate calibration domain, correspondence count, and scene content. Finally, SH degree zero in the main study and full-image DTU appearance evaluation restrict comparisons with other reconstruction settings.

8 Conclusion

Depth priors can substantially improve sparse-view 3DGS, but their value depends on initialization, supervision, view coverage, and calibration. Clean dense-reference initialization improves both PSNR and reference-depth agreement, whereas corrupted priors can erase or reverse these gains. Coherent affine errors are especially damaging in the tested configurations, and selected experiments demonstrate that PSNR and geometry can move in opposite directions. Depth Anything V2 provides a useful empirical test of these issues, while DTU confirms that depth information can improve independently measured real-scene geometry. A reliable evaluation should therefore report calibration-sensitive geometry and multiple appearance metrics, with conclusions tied to matched experimental conditions rather than a universal threshold or loss weight.

References

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4), 2023.
- [2] Y. Foroutan, D. Rebain, K. M. Yi, and A. Tagliasacchi. Evaluating Alternatives to SFM Point Cloud Initialization for Gaussian Splatting. arXiv:2404.12547, 2024.
- [3] J. Li, J. Zhang, X. Bai, J. Zheng, X. Ning, J. Zhou, and L. Gu. DNGaussian: Optimizing Sparse-View 3D Gaussian Radiance Fields with Global-Local Depth Normalization. *CVPR*, 2024.
- [4] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao. Depth Anything V2. *NeurIPS*, 2024.
- [5] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. *ECCV*, 2020.
- [6] R. Jensen, A. Dahl, G. Vogiatzis, E. Tola, and H. Aanæs. Large Scale Multi-view Stereopsis Evaluation. *CVPR*, 2014.
- [7] V. Ye, R. Li, J. Kerr, M. Turkulainen, B. Yi, Z. Pan, O. Seiskari, J. Ye, J. Hu, M. Tancik, and A. Kanazawa. gsplat: An Open-Source Library for Gaussian Splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025.